

Project : Data Analyst

IT Company

— by Mohamad Ikhsan Nurulloh —

Assignment

1. How is the data quality of the excel data provided, and what can be done to improve it?
2. Cleanse the data and describe the phases. (you can attach files from Jupyter Notebook, Google Colab, or something similar)
3. Share your insights from the data.

[Dataset Link](#)

Dataset Preview (Sewa ATM)

Daftar Sewa ATM

NO	Tanggal Request	Nama LOKASI	KOTA	MASA SEWA (TAHUN)	Delivery	Type	Jatuh tempo lag 3-4th	Tagihan			Tagihan	
								I			2	
								No. Inv.	Period	JT48	No. Inv.	Period
1	21 September 2010	PLN Kota	Jkt-HUB II RPC		7-Oct-10	522	Oct-13	379-11-10	Oct-10		379-11-10	Nov-10
2	21 September 2010	Perumnas	Jkt-HUB I RPC		7-Oct-10	522	Oct-13	379-11-10	Oct-10		379-11-10	Nov-10
3	21 September 2010	Tip Top Depok	Jkt-HUB VIII RPC		7-Oct-10	522	Oct-13	379-11-10	Oct-10		379-11-10	Nov-10
4	21 September 2010	Tip Top Pdk Bambu	Jkt-HUB VI RPC		7-Oct-10	522	Oct-13	379-11-10	Oct-10		379-11-10	Nov-10
5	21 September 2010	Tip Top Rawamangun	JKT-HUB VII		7-Oct-10	529EPP7	Oct-13	379-11-10	Oct-10		379-11-10	Nov-10
6	21 September 2010	LIA Pramuka	Jkt-HUB I RPC		7-Oct-10	522	Oct-13	379-11-10	Oct-10		379-11-10	Nov-10
7	21 September 2010	PLN Pdk Gede	Jkt-HUB VI RPC		8-Oct-10	522	Oct-13	379-11-10	Oct-10		379-11-10	Nov-10
8	21 September 2010	Esa Unggul	Jkt-HUB II RPC		11-Oct-10	522	Oct-13	379-11-10	Oct-10		379-11-10	Nov-10
9	21 September 2010	RSPAD Kartika OFF DES 17	Jkt-HUB IV OFF		11-Oct-10	522	Oct-13	379-11-10	Oct-10		379-11-10	Nov-10
10	21 September 2010	PLN Grogol, JT OFF JAN 2018	Jkt-HUB II OFF		13-Oct-10	522	Oct-13	379-11-10	Oct-10		379-11-10	Nov-10
11	21 September 2010	KK Meruya	Jkt-HUB II RPC		15-Oct-10	522	Oct-13	379-11-10	Oct-10		379-11-10	Nov-10
12	21 September 2010	Lippo Karawaci / PLN Bekasi/RSMI	Jkt-HUB VI OFF		3-Oct-11	522	Oct-13	411-12-10	Oct-10		411-12-10	Nov-10
13	21 September 2010	KK RSPAD / PLN Menteng	Jkt-HUB IV RPC		21-Oct-10	522	Oct-13	411-12-10	Oct-10		411-12-10	Nov-10
14	28 September 2010	Capem Setiabudi	Bandung OFF		3-Dec-10	522	Jan-14	014-01-11	Jan-11		034-02-11	Feb-11
15	28 September 2010	KCP Cimahi	Bandung OFF		3-Dec-10	522S	Jan-14	014-01-11	Jan-11		034-02-11	Feb-11
16	28 September 2010	ADA Majapahit / CAPEM MAJAP.	Semarang RPC		8-Dec-10	522	Jan-14	014-01-11	Jan-11		034-02-11	Feb-11
17	28 September 2010	Citraland Plaza	Semarang OFF		8-Dec-10	522	Jan-14	014-01-11	Jan-11		034-02-11	Feb-11
18	28 September 2010	KCP Srandol	Semarang RPC		8-Dec-10	522	Jan-14	014-01-11	Jan-11		034-02-11	Feb-11
19	28 September 2010	Mall Rengas	Glasco RPC		8-Dec-10	522	Dec-13	014-01-11	Jan-11		034-02-11	Feb-11

Dataset Preview (Aktivitas Perawatan ATM)

Data aktivitas ATM										
No.	Tanggal	Nomor	ATM ID	Lokasi	Jenis Layat	Laporan Kerusakan	Pekerjaan	Nomor Tiket	Penggantian Sparepart	Jumlah Par Status
1	16 Juli 2020	88826	ATM00001	RUANG UAT	Error	Hardware	Check & Clez	2007-100	Hard Disk	1
2	03 Maret 202	86010	ATM1200290	PLN AHMAD	Error	Hardware	Check & Clez	2003-028	CCA Dispenser (CCA Kit Presenter Ctrl.) 49-74724 EPP5(BSC),LGE_STTL,ENG(AU),QZ1,BLANK,X,I,C	1
3	02 Septembe	90681	ATM1200440	GRAHA ENE	PM 1	Preventive Maintenance	Check & Clez			
4	09 Septembe	89466	ATM120047014		Installation	Installation	Instalasi ATM			
5	29 Maret 202	87602	ATM2000200	BKP MEDAN	PM pertama	Preventive Maintenance	Check & Clez			
6	27 April 2020	87445	ATM2000200	BKP MEDAN	PM pertama	Preventive Maintenance	Check & Clez			
7	24 November	89385	ATM2000200	BKP MEDAN	PM pertama	Preventive Maintenance	Check & Clez			
8	24 Agustus 2	87409	ATM2000200	BKP MEDAN	PM pertama	Preventive Maintenance	Check & Clez			
9	22 Juni 2020	87425	ATM2000200	BKP MEDAN	PM pertama	Preventive Maintenance	Check & Clez			
10	21 Septembe	87604	ATM2000200	BKP MEDAN	PM pertama	Preventive Maintenance	Check & Clez			
11	27 Juli 2020	87417	ATM2000200	BKP MEDAN	PM pertama	Preventive Maintenance	Check & Clez			
12	26 Mei 2020	87608	ATM2000200	BKP MEDAN	PM pertama	Preventive Maintenance	Check & Clez			
13	22 Oktober 2	89388	ATM2000200	BKP MEDAN	PM pertama	Preventive Maintenance	Check & Clez			
14	27 Januari 20	87621	ATM2000200	BKP MEDAN	PM pertama	Preventive Maintenance	Check & Clez			
15	30 Maret 202	87606	ATM2000200	BKP AR HAK	PM pertama	Preventive Maintenance	Check & Clez			
16	25 November	89384	ATM2000200	BKP AR HAK	PM pertama	Preventive Maintenance	Check & Clez			
17	28 April 2020	87439	ATM2000200	BKP AR HAK	PM pertama	Preventive Maintenance	Check & Clez			
18	16 Desember	89376	ATM2000200	BKP AR HAK	PM pertama	Preventive Maintenance	Check & Clez			
19	24 Septembe	87403	ATM2000200	BKP AR HAK	PM pertama	Preventive Maintenance	Check & Clez			
20	23 Juni 2020	87424	ATM2000200	BKP AR HAK	PM pertama	Preventive Maintenance	Check & Clez			
21	19 Juli 2020	87418	ATM2000200	BKP AR HAK	PM pertama	Hardware	Check & Clez			
22	27 Mei 2020	87438	ATM2000200	BKP AR HAK	PM pertama	Preventive Maintenance	Check & Clez			
23	23 Oktober 2	89390	ATM2000200	BKP AR HAK	PM pertama	Preventive Maintenance	Check & Clez			
24	28 Januari 20	87623	ATM2000200	BKP AR HAK	PM pertama	Preventive Maintenance	Check & Clez			
25	23 November	89383	ATM2000200	KCP BINJAI	PM pertama	Preventive Maintenance	Check & Clez			
26	29 April 2020	87440	ATM2000200	KCP BINJAI	PM pertama	Preventive Maintenance	Check & Clez			

Assignment 1

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Data Quality

1. High bad data quality means that the information in the cell doesn't match what it should be (i.e., it's not accurate). This could be due to human error or it could be because the source of your data is unreliable.
2. Bad data format means that the data in the cell isn't formatted correctly for Excel. For example, in your dataset given there is a column was supposed to hold dates, but instead held numbers instead, then that would be considered bad data format.
3. Many missing values mean that cells don't contain any information at all—they just have a blank space where something should be written down.

Improve Data Quality

1. Use validation. This will help you find and fix errors in your Excel spreadsheets before they cause problems. Create a list of acceptable values for each cell in your spreadsheet. This can help prevent typos or data entry errors from making it into your dataset.
2. Using column formatting to set the data type of your columns. You can also use conditional formatting to highlight cells that contain errors or other anomalies.
3. Last, make sure you have clean data. This means a few things: make sure your data is accurate (and can be verified), make sure that it's consistent across all the fields you're using, and make sure there aren't any missing values in your data set.

Assignment 2

— by Mohamad Ikhsan Nurulloh —

Data Cleansing Steps (Sewa ATM)

1. Importing pandas and numpy
2. Mounting dataset
3. Dropping column with NaN Value
4. Transforming data frame to friendly database format
5. Formatting empty column and inconsistent value to NaN
6. Setting column data type to python format
7. Formatting empty column and inconsistent value
8. Removing unnecessary space

Dataset Preview After Cleansing(Sewa ATM)

tanggal_request	nama_lokasi	kota	masa_sewa	delivery	tipe	jatuh_tempo	no_invoice	periode	tagihan	keterangan
21/09/2010	PLN Kota	Jakarta Hub II	1	07/10/2010	522	01/10/2013	379-11-10	01/10/2010	1	rpc
21/09/2010	Perumnas	Jakarta Hub I	1	07/10/2010	522	01/10/2013	379-11-10	01/10/2010	1	rpc
21/09/2010	Tip Top Depok	Jakarta Hub VIII	1	07/10/2010	522	01/10/2013	379-11-10	01/10/2010	1	rpc
21/09/2010	Tip Top Pdk Bambu	Jakarta Hub VI	1	07/10/2010	522	01/10/2013	379-11-10	01/10/2010	1	rpc
21/09/2010	Tip Top Rawamangun	Jakarta Hub VII	1	07/10/2010	529EPP7	01/10/2013	379-11-10	01/10/2010	1	aktif
21/09/2010	LIA Pramuka	Jakarta Hub I	1	07/10/2010	522	01/10/2013	379-11-10	01/10/2010	1	rpc
21/09/2010	PLN Pdk Gede	Jakarta Hub VI	1	08/10/2010	522	01/10/2013	379-11-10	01/10/2010	1	rpc
21/09/2010	Esa Unggul	Jakarta Hub II	1	11/10/2010	522	01/10/2013	379-11-10	01/10/2010	1	rpc
21/09/2010	RSPAD Kartika DES 17	Jakarta Hub IV	1	11/10/2010	522	01/10/2013	379-11-10	01/10/2010	1	off
21/09/2010	PLN Grogol, JT JAN 2018	Jakarta Hub II	1	13/10/2010	522	01/10/2013	379-11-10	01/10/2010	1	off
21/09/2010	KK Meruya	Jakarta Hub II	1	15/10/2010	522	01/10/2013	379-11-10	01/10/2010	1	rpc
21/09/2010	Lippo Karawaci / PLN Bekasi/RSMK Cibubur	Jakarta Hub VI	1	03/10/2011	522	01/10/2013	411-12-10	01/10/2010	1	off
21/09/2010	KK RSPAD / PLN Menteng	Jakarta Hub IV	1	21/10/2010	522	01/10/2013	411-12-10	01/10/2010	1	rpc
28/09/2010	Capem Setiabudi	Bandung	1	03/12/2010	522	01/01/2014	014-01-11	01/01/2011	1	off
28/09/2010	KCP Cimahi	Bandung	1	03/12/2010	522S	01/01/2014	014-01-11	01/01/2011	1	off
28/09/2010	ADA Majapahit / CAPEM MAJAPAHIT	Semarang	1	08/12/2010	522	01/01/2014	014-01-11	01/01/2011	1	rpc
28/09/2010	Citraland Plaza	Semarang	1	08/12/2010	522	01/01/2014	014-01-11	01/01/2011	1	off
28/09/2010	KCP Srandol	Semarang	1	08/12/2010	522	01/01/2014	014-01-11	01/01/2011	1	rpc
28/09/2010	Mall Ramayana	Cilegon	1	08/12/2010	522	01/12/2013	014-01-11	01/01/2011	1	rpc
28/09/2010	Kios Batu Malang	Malang	1	21/12/2010	522	01/01/2014	014-01-11	01/01/2011	1	rpc
28/09/2010	KK Blimbing / KCU Malang	Malang	1	21/12/2010	522	01/01/2014	014-01-11	01/01/2011	1	rpc
28/09/2010	KCP Bondowoso	Jember	1	21/12/2010	522	01/04/2014	138-05-11	01/04/2011	1	rpc
28/09/2010	Probolinggo	Probolinggo	1	21/12/2010	522	01/01/2014	014-01-11	01/01/2011	1	rpc
28/09/2010	Pasuruan	Probolinggo	1	21/12/2010	522	01/01/2014	014-01-11	01/01/2011	1	rpc
28/09/2010	Dolog UPG	Makasar	1	21/12/2010	522	01/01/2014	014-01-11	01/01/2011	1	rpc
28/09/2010	Bondowoso / KAMPUS DIP SOLODONG	Makasar	1	21/12/2010	522	01/01/2014	014-01-11	01/01/2011	1	rpc

Data Cleansing Steps (Perawatan ATM)

1. Importing pandas and numpy
2. Mounting dataset
3. Dropping column with NaN Value
4. Converting month name to english
5. Dropping unnecessary column
6. Setting column data type to python format
7. Formatting empty column and inconsistent value
8. Removing unnecessary space

Dataset Preview After Cleansing (Aktivitas Perawatan ATM)

no	tanggal	nomor ATM_ID	lokasi	jenis_layanan	laporan_kerusakan	pekerjaan_dilakukan	penggantian_sparepart	jumlah_part	CSE
1	16/07/2020	88826 ATM00001	Ruang Uat Dpti	Error	Hardware	Check & Clean all devices, replace	Hard Disk	1	M. Yani
2	03/03/2020	86010 ATM120029008	Pln Ahmad Yani	Error	Hardware	Check & Clean all devices, replace EPI CCA Dispenser (CCA Kit Present)		1	Hendri Apridzal
3	02/09/2020	90681 ATM120044005	Graha Energy	PM 1	Preventive Maintenance	Check & Clean all devices, no replace	Tanpa Penggantian Sparepart	0	Ozi Medina
5	29/03/2020	87602 ATM20002001	Bkp Medan	PM 1	Preventive Maintenance	Check & Clean all devices, no replace	Tanpa Penggantian Sparepart	0	Rajamin Nasution
6	27/04/2020	87445 ATM20002001	Bkp Medan	PM 1	Preventive Maintenance	Check & Clean all devices, no replace	Tanpa Penggantian Sparepart	0	Rajamin Nasution
7	24/11/2020	89385 ATM20002001	Bkp Medan	PM 1	Preventive Maintenance	Check & Clean all devices, no replace	Tanpa Penggantian Sparepart	0	Rajamin Nasution
8	24/08/2020	87409 ATM20002001	Bkp Medan	PM 1	Preventive Maintenance	Check & Clean all devices, no replace	Tanpa Penggantian Sparepart	0	Rajamin Nasution
9	22/06/2020	87425 ATM20002001	Bkp Medan	PM 1	Preventive Maintenance	Check & Clean all devices, no replace	Tanpa Penggantian Sparepart	0	Rajamin Nasution
10	21/09/2020	87604 ATM20002001	Bkp Medan	PM 1	Preventive Maintenance	Check & Clean all devices, no replace	Tanpa Penggantian Sparepart	0	Rajamin Nasution
11	27/07/2020	87417 ATM20002001	Bkp Medan	PM 1	Preventive Maintenance	Check & Clean all devices, no replace	Tanpa Penggantian Sparepart	0	Rajamin Nasution
12	26/05/2020	87608 ATM20002001	Bkp Medan	PM 1	Preventive Maintenance	Check & Clean all devices, no replace	Tanpa Penggantian Sparepart	0	Rajamin Nasution
13	22/10/2020	89388 ATM20002001	Bkp Medan	PM 1	Preventive Maintenance	Check & Clean all devices, no replace	Tanpa Penggantian Sparepart	0	Rajamin Nasution
14	27/01/2020	87621 ATM20002001	Bkp Medan	PM 1	Preventive Maintenance	Check & Clean all devices, no replace	Tanpa Penggantian Sparepart	0	Rajamin Nasution
15	30/03/2020	87606 ATM20002002	Bkp Ar Hakim Mdn	PM 1	Preventive Maintenance	Check & Clean all devices, no replace	Tanpa Penggantian Sparepart	0	Rajamin Nasution
16	25/11/2020	89384 ATM20002002	Bkp Ar Hakim Mdn	PM 1	Other	Check & Clean all devices, no replace	Tanpa Penggantian Sparepart	0	Rajamin Nasution
17	28/04/2020	87439 ATM20002002	Bkp Ar Hakim Mdn	PM 1	Preventive Maintenance	Check & Clean all devices, no replace	Tanpa Penggantian Sparepart	0	Rajamin Nasution
18	16/12/2020	89376 ATM20002002	Bkp Ar Hakim Mdn	PM 1	Preventive Maintenance	Check & Clean all devices, no replace	Tanpa Penggantian Sparepart	0	Rajamin Nasution
19	24/09/2020	87403 ATM20002002	Bkp Ar Hakim Mdn	PM 1	Preventive Maintenance	Check & Clean all devices, no replace	Tanpa Penggantian Sparepart	0	Rajamin Nasution
20	23/06/2020	87424 ATM20002002	Bkp Ar Hakim Mdn	PM 1	Preventive Maintenance	Check & Clean all devices, no replace	Tanpa Penggantian Sparepart	0	Rajamin Nasution
21	19/07/2020	87418 ATM20002002	Bkp Ar Hakim Mdn	PM 1	Hardware	Check & Clean all devices, no replace	Tanpa Penggantian Sparepart	0	Rajamin Nasution
22	27/05/2020	87438 ATM20002002	Bkp Ar Hakim Mdn	PM 1	Preventive Maintenance	Check & Clean all devices, no replace	Tanpa Penggantian Sparepart	0	Rajamin Nasution
23	23/10/2020	89390 ATM20002002	Bkp Ar Hakim Mdn	PM 1	Other	Check & Clean all devices, no replace	Tanpa Penggantian Sparepart	0	Rajamin Nasution
24	28/01/2020	87623 ATM20002002	Bkp Ar Hakim Mdn	PM 1	Preventive Maintenance	Check & Clean all devices, no replace	Tanpa Penggantian Sparepart	0	Rajamin Nasution
25	23/11/2020	89383 ATM20002003	Kcp Binjai	PM 1	Preventive Maintenance	Check & Clean all devices, no replace	Tanpa Penggantian Sparepart	0	Rajamin Nasution
26	29/04/2020	87440 ATM20002003	Kcp Binjai	PM 1	Preventive Maintenance	Check & Clean all devices, no replace	Tanpa Penggantian Sparepart	0	Rajamin Nasution
27	02/12/2020	89375 ATM20002002	Kcp Binjai	PM 1	Preventive Maintenance	Check & Clean all devices, no replace	Tanpa Penggantian Sparepart	0	Rajamin Nasution

Data Cleansing



Google Colab Link

1. [Sewa ATM](#)
2. [Perawatan ATM](#)

Link for the Data After Cleansing

<https://drive.google.com/drive/folders/11YSI7MqhWiTqVzfxdFb8kSs48Scuj-Sv?usp=sharing>

Assignment 3

— by Mohamad Ikhsan Nurulloh —

Data Insights (Sewa ATM)

1. 57 cities and 847 locations that are rented out by 40,673 renters. There are also 487 invoices that have been issued by these rentees.
2. Surabaya has the highest rental request number by city at 2,290 rentals, while Tip Top Rawamangun has the highest rental request number by location at 100 rentals.
3. The highest rental request is between Q2 2015 with 2,368 rentals. The most rentals ATM is 529EPP7CL type with 10,774 rentals.

Data Insights (Perawatan ATM)

1. The number of ATMs in the world is 717. In the past year, there were 3026 repair activities on ATMs. The number of CSE is 22.
2. The highest repair request is between November 2020 with 13 request and the lowest repair request is between March 2020 with 2 requests.
3. The highest number of broken ATMs is caused by 'installiation' with 72,9% request and the highest number of services ATMs are PM 1 by 1,992 request.
4. M. Yani is the CSE with the highest handled service who handling 308 services and KCU Banjarmasin 2 is the highest location with 14 service requests.

Data Insights

IT Company



Google Data Studio Interactive Dashboard Link

<https://datastudio.google.com/reporting/b36a00c4-e22d-4811-be05-ca1819b118bd>

Click Here For Others Portfolio

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